MILAD RABIEI

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Education

Doctorate, Robotics *2026 - present*

Human-Robot Interfaces and Interaction (HRI²), Italian Institute of Technology

Master of Science, Robotics Engineering *2024 - 2026*

DIBRIS, Università degli Studi di Genova, 110/110 with honours
Thesis: Intra-task generalization via semantic generalization of motion strategies
Advised by Dr. Elena Merlo and Dr. Marta Lagomarsino,
Supervised by Prof. Enrico Simetti and and Dr. Arash Ajoudani

Bachelor of Science, Electrical Engineering *2019 - 24*

ECE Dept., Shahid Beheshti University, Iran
Thesis: Landmark tracking and object detection, complete vision system for self-driving cars
Supervised by Prof. Mohammad Hossein Moaiyeri

Publications

Mutual Impact of Feature Selection and Privacy-preserving Mechanisms

International Journal of Machine Learning and Cybernetics, Springer, 2025
M Alishahi, V Moghtadaiee, A Fathalizadeh, **Milad Rabiei**

Differentially Private GANs for Generating Synthetic Indoor Location Data

International Journal of Information Security, Springer, 2025
V Moghtadaiee, M Alishahi, **Milad Rabiei**

A Density-based Lane Keeping System for A Cost-efficient Autonomous Vehicle Platform

Manuscript, Arxiv, 2024
Younesi*, F., **Milad Rabiei***, Keivanfard*, S., Sharifi*, M., GhayourNajafabadi, M., Moadeli, B., Jafari, A., Moaiyeri, M. H.

Academic Experiences

Guest Student - Human-robot interfaces and interaction (HRI²), IIT *2026*

– Thesis work on imitation learning for robot manipulation

Research Assistant - Cyberspace research institute, SBU *2023 - 24*

– Generative networks, private data analysis, differential privacy, federated learning

Research Assistant - Robotics and intelligent automation lab, SBU *2020 - 23*

– Vision for AGVs (level 4), segmentation, detection, and localization

Lead Teaching Assistant - ECE department, SBU *2021 - 24*

- Machine Learning (graduate-level), Prof.: Reza Ghaderi
- Intro to Artificial Intelligence (graduate-level), Prof.: Atefeh Aghaei
- Computer Programming, Prof.: Vahideh Moghtadaiee

Skills

Technical: Python (OpenCV, PyTorch, TF/Keras, etc.), C/C++, MATLAB, Mujoco, PDDL/PDDL+, FastDDS, ROS/ROS 2, Gazebo, SAM, VLMs (Llama3, OpenAI API)
Embedded: Nvidia Jetson (CUDA), Raspberry Pi, dsPIC, NodeMCU ESP8266, ESP32
Languages: English (proficient), Italian (elementary), Persian (native)

Honors and Awards

(2023) Recipient of MCI R&D research grant for small-size autonomous vehicle
(2022) Recipient of honorary Dean and Departmental research grants for Robotics Lab at SBU
(2022) 1st place, Individual team section, Autonomous vehicles league, IranOpen RoboCup 2022
(2022) 1st place, Technical challenge, Autonomous vehicles league, IranOpen RoboCup 2022
(2021) 2nd place, Race section, Autonomous cars simulation, Fira RoboWorld Cup SDE
(2021) 1st place, Race section, Autonomous cars simulation, Iran Fira RoboWorld Cup
(2021) 2nd place, Urban section, Autonomous cars simulation, Iran Fira RoboWorld Cup
(2019) Top 99th percentile in national university entrance exam among 140K participants

Other Experiences

(2026-) Junior reviewer (Young Reviewers Program), IEEE Robotics and Automation Society
(2024) Member of National Committee, IranOpen RoboCup International Competitions
(2023) Member of Technical Committee, IranOpen RoboCup International Competitions
(2023) Machine Learning Course Instructor, Iran RoboCamp, Tehran
(2023) Organizer of Deep Learning Reading Group, SBU ECE Department
(2022) Network Programming Intern, Amin Optical Communications Technologies (FANA), Tehran

Conference Coordinator:
(2025) The 19th International Conference on Intelligent Autonomous Systems (IAS19), Genoa, Italy.
(2022) The 4th Iranian International Conference on Microelectronics (IICM), Tehran, Iran
(2022) The 13th Power Electronics and Drives: Systems and Technologies Conference (PEDSTC), Tehran, Iran